

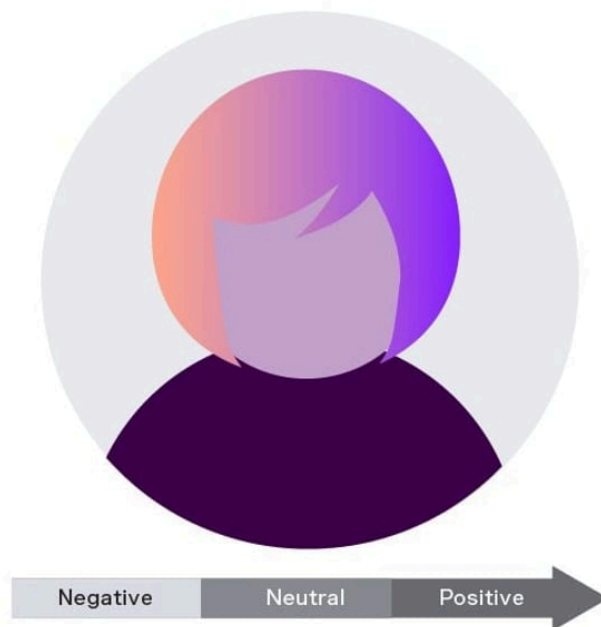


What Is Sentiment Analysis? How Deeper Insight Leads to Customer Satisfaction

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With so much customer data now available, the biggest challenge for organizations is having the time to analyze it and extract the valuable insights it holds. That's where **sentiment analysis** comes in, allowing businesses to evaluate data at scale to reveal exactly what their customers are feeling.

In this article, we'll define sentiment analysis, explore its benefits, and show you how to perform sentiment analysis and use the results for maximum customer satisfaction.





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Let's start with a quick sentiment analysis definition. Sentiment analysis is the process of analyzing written text and sometimes spoken words to identify their emotional tone, whether it's positive, negative, or neutral.

The process involves data mining, machine learning (ML), and artificial intelligence (AI), and uses a scoring mechanism to quantify emotions. It's also sometimes known as opinion mining because organizations can use it to gauge customer opinions on their products and services.

Sentiment analysis tools can scan large volumes of text data like emails, social media comments, survey responses, and reviews. These tools can also determine emotions in live conversations, as well as spoken words transcribed into text.

Why is sentiment analysis important?

Sentiment analysis helps companies to understand their customers, which leads to a better customer experience and increased loyalty. Here's why:

It's objective

Humans come with built-in biases — we can't help it. AI sentiment analysis tools can scan and classify text objectively, avoiding the personal bias that creeps into manual evaluation. This gives organizations more consistent results.

Let's say a customer says: "I love shopping at this store (although it's a little on the expensive side)." A human reviewer might decide to classify this sentence as positive. A sentiment analysis tool would detect both the opinions in the sentence and classify them as neutral.

It helps you analyze at scale

To reveal what customers think, businesses have to wade through a ton of unstructured data. This can be overwhelming and takes up a lot of resources, especially as companies grow and generate more data.

With sentiment analysis, the automated tools allow you to analyze emotions at scale, quickly and efficiently, and at a lower cost. Employees can focus on other



The speed of automated sentiment analysis enables you to gain insights from customer data in real time — which means you can respond quickly to a negative experience, a potential PR crisis, or a new market trend.

For example, if the analysis reveals detrimental comments about your business on social media, you can immediately address the situation and avoid reputational damage. You can even configure these tools to alert you if they spot negative sentiment for specific keywords.

What are sentiment analysis use cases?

The emotion detection and recognition market size is expected to reach \$74.80 billion by 2029, but how are companies using it to their advantage?

Improve customer support

Sentiment analysis in contact centers enables you to spot customer issues that deserve urgent attention, prioritize them, and direct them to the right people. It will also detect customer reviews that mention problems with particular product features and rank topics by level of urgency.

You can identify and address common pain points, and personalize your responses based on the customer's mood. Automation also makes it easy to gather feedback across all channels, which helps customer support teams respond faster.

Enhance products and services

The ability to extract specific insights from customer feedback helps you to adjust products and services accordingly. Based on the amount of positive and negative sentiment, you can find out which product features people love the most and which ones they find glitchy or frustrating.

You can pass the feedback to the product development team to make targeted improvements and also use sentiment analysis to monitor whether customers are seeing an uptick in functionality with the new versions.



by analyzing mentions of your brand on social media, forums, blogs, and in the news. When you evaluate the prevailing attitude of your customers, you'll be better placed to manage your reputation.

Track campaign performance

You can check the impact of a PR campaign or the launch of a new product on overall brand sentiment. For marketing campaigns, sentiment analysis also allows you to track conversations for customer attitudes. If a campaign hasn't generated the expected response, you can tweak it based on real-time data.

Conduct market research

Sentiment analysis helps you research customer opinions across the whole market, not just for your own products and services. For instance, you can analyze online review sites, social media posts, and the results of customer surveys to gain insights, spot trends, and identify new opportunities. This includes assessing how people view you compared to your competitors.

Sentiment analysis examples

As the technology becomes more accessible, there are plenty of real-world examples of sentiment analysis. Here are a few:

To evaluate and adjust a marketing campaign

Nike uses social listening and sentiment analysis during marketing campaigns, monitoring what their millions of social media followers are saying. In the 30th-anniversary "Just Do It" campaign with Colin Kaepernick, Nike knew the strategy would be controversial, so they used sentiment analysis to their advantage.

They deployed a dedicated team to respond quickly on social media, adjusted their messaging based on comments, and partnered with influencers as backup. The strategy paid off, with a huge increase in brand mentions as well as earned media — plus a 67% approval rating among young customers.

For proactive customer service



flags it for attention. This helps the customer service team reach out to customers who may have had negative experiences, offering refunds or replacements to ensure satisfaction and prevent further complaints.

To manage customer frustration

Delta Airlines uses sentiment analysis on social media and customer feedback to monitor real-time reactions during flight delays or cancellations. If a customer tweets frustration, using language like "disappointed" or "never flying with Delta again," sentiment analysis flags these messages. The customer service team can then respond immediately, offering solutions or compensation to prevent escalation and maintain customer loyalty.



Sentiment analysis :

Depending on your needs, you can choose between various types of sentiment analysis. Most of these help gauge “polarity”, which is the overall feeling conveyed by a particular text, phrase, or word.

Fine-grained scoring

This type of sentiment analysis categorizes emotion into multiple levels, similar to the 5-star rating system used by consumer websites. Based on polarity, it rates user sentiment on a scale of 0 to 100, with equal segments representing very positive, positive, neutral, negative, and very negative.

You can split the sentiment further into specific emotions, such as happy, excited, or impressed for positive sentiment. Fine-grained scoring is also useful for processing comparative expressions.

Aspect-based

Aspect-based sentiment analysis (ABSA) focuses on specific aspects of a product or service, and correlates customer intent with keywords relating to that aspect. This means you can narrow your focus when evaluating a body of text or a conversation and analyze the aspect in detail, providing a deeper understanding of a product's success or failure.

Intent-based

Intent-based sentiment analysis uncovers the intent behind the words, showing if a customer is likely to make a purchase or leave the company. Marketers use it during the sales cycle to determine a prospect or customer's level of interest and to recommend relevant products, while support teams can identify and reach out to those at risk of churn.

Emotional detection

This method is a more complex discipline of sentiment analysis, as it's all about interpreting the psychological and emotional state of the customer. Instead of categorization using polarities (positive, negative, neutral), it picks out specific emotions conveyed by their words — such as joy, anger, frustration, sadness, or indifference.



As mentioned, sentiment analysis harnesses natural language processing (NLP) and ML. These technologies teach computer software to analyze and interpret text and speech in a human-like way. Here's a basic outline of how it works:

Preprocessing

Preprocessing is about cleaning and standardizing the data so that the tools can understand the main message. This involves the following:

- Removing "stop words" (like at, of, and for), which aren't important to the meaning of the sentence
- Removing irrelevant data, such as HTML tags and special characters
- Completing tokenization, which is the process of breaking a sentence down into several elements or "tokens"
- Using lemmatization or stemming to convert words into their root form (for example, the root form of am is be)

Keyword analysis

At the next stage, the tool analyzes the extracted keywords using NLP and gives each one a sentiment score, providing a relative perception of the emotion being expressed. The scores are assigned using a measurement scale from 0 to 10, with 10 representing complete satisfaction and 0 representing extreme disappointment.

Sentiment analysis techniques

Now we'll look at the different techniques you can employ to carry out the sentiment analysis processes.

Lexicon-based analysis

This is a rule-based approach in which the software is trained to detect, classify, and score certain keywords. First, you create groups of words called



The software looks for words that fit into these groups and generates a sentiment score for each category. It's possible to set up special rules to correctly identify double negatives (for example, not great should be classified as a negative sentiment).

Machine learning-based analysis

This technique uses ML techniques and classification algorithms, such as neural networks and deep learning. As well as analyzing the words themselves, the software learns that the order of the words may affect the sentiment score.

It involves creating a sentiment analysis model and feeding it with large numbers of diverse examples from sentiment analysis datasets. Models trained repeatedly on known data will eventually predict the sentiment in unknown data.

Hybrid

The hybrid approach combines lexicon-based and ML-based analysis, taking the best bits of each system to optimize speed and accuracy. For instance, a lexicon-based approach generates near-instant results, but ML-based analysis is flexible enough to handle more complex scenarios. However, it is worth noting that hybrid analysis requires more resources.

How to do sentiment analysis

Wondering exactly how to conduct sentiment analysis? Here's a simple step-by-step guide:

1. Data collection

The first step is to collect the data that you want to analyze. You might evaluate comments from a certain segment of your audience or conversations with an individual customer. Typically, data collection is done with a web scraping bot or a scraping [application programming interface \(API\)](#).

2. Data preprocessing



3. Choose a sentiment analysis approach

Now you have a decision to make — will you pick a lexicon-based, ML-based, or hybrid approach? Think carefully about the pros and cons of each one, factoring in the amount of sentiment analysis data you have and what you want to learn.

4. Implementation

You'll also need to choose between fine-grained, aspect-based, intent-based, and emotional detection analysis. In some cases, you can use a combination. Make sure your sentiment analysis tools integrate with existing systems, such as customer support software.

5. Model training and evaluation (if using machine learning)

If you're going down the ML route, it's time to train your model and check its results. You'll need to preprocess a dedicated and classified sentiment analysis dataset and label it manually. This labeled data will be used to train the model by comparing correctly classified data with an incorrectly classified version.

6. Analysis and interpretation

Once you're happy the model's output is accurate, you can use it to analyze fresh data. Sentiment analysis tools typically generate visual reports to help you interpret the results. You can also view the findings at document level, sentence level, and sub-sentence level.

What are the challenges in sentiment analysis?

Although machines are pretty smart, they can't always understand the nuances of human language — such as subjectivity. They also have trouble distinguishing real reviews from fake ones created by bots. Here are some other challenges in sentiment analysis:



know the original context of the question itself. For example, if the customer just answers, "The color," it would be impossible to classify this as positive or negative without knowing whether the question was, "What did you like about this sweater?" or "What did you dislike about it?"

Carrying out pre or post processing to establish context is time-consuming. The Vonage Conversation Analyzer makes this easier as the context from your interactions is automatically stored with your customer data.

Sarcasm and irony

If a sentence contains sarcasm or irony, it's challenging for a computer to interpret the sentiment correctly. For instance, if a customer review for an airline says, "And just to top off a perfect experience, my plane was late." Most machines would not understand that the writer was being sarcastic, and they'd interpret the word perfect as positive sentiment.

Negation

Negation alters the meaning of a sentence by adding a negative word. For example, "The sweater was good quality" and "The sweater was not good quality." This is tricky for sentiment analysis tools to understand, because most of the words are the same.

It's even harder if the negation happens across two sentences. For instance, "I thought the sweater would be good quality. It wasn't." Meanwhile, double negatives ("I won't never use this company again!") are next to impossible for machines to decipher.

Multipolarity

Sentiment analysis tools also face difficulties if a sentence contains more than one sentiment. For instance, "I love the color, but the quality is poor". The only way for the computer to interpret multipolarity correctly is to use aspect-based sentiment analysis, looking at each section separately to understand its emotion.

Idioms



miss valuable insight.

Evolving language

Human language is always in flux, with new abbreviations, acronyms, and even words popping into our usage. The meaning of words can also change over time ("woke" being a classic example). It's hard for sentiment analysis models to keep track, especially when people also use incorrect spelling and grammar.

Sentiment vs. semantic analysis

Both linguistic technologies can be deployed to help businesses understand their customers better. Both use ML and NLP. Let's compare the differences:



Sentiment Analysis

Aims to understand underlying emotions and emotional tone

Categorizes words and phrases as positive, negative, or neutral

Based on specific datasets and rule-based lexicons

Uses tokenization and sentiment scoring

Semantic Analysis

Aims to understand the meaning and context behind the words

Analyzes language patterns to make sense of the relationship between words, phrases, and concepts

Works with more extensive and diverse information

Uses part-of-speech tagging, entity recognition, and semantic parsing



By showing you how customers feel, sentiment analysis gives you the chance to take action and improve customer experience (CX) for your business. Here are the key sentiment analysis applications for enhancing CX:

- Gather instant feedback and respond quickly
- Prioritize urgent support requests
- Identify and address friction points at touchpoints in the customer journey
- Identify the things they love about you — and keep doing them
- Personalize responses to match the customer's mood
- Target customers with relevant messaging
- Be proactive in anticipating customer needs
- Use feedback to improve products and services
- Ensure compliance (such as checking that agents always ask for consent to record calls)

Harness the power of sentiment analysis — and understand your customers better

Sentiment analysis provides deep insights into the minds of your customers, giving you the data you need to improve their experiences. With speech analytics alongside speech-to-text and transcription capabilities for text-based analysis, that's exactly what Vonage delivers.

Real-time analysis lets you take quick action, such as a supervisor stepping in to rescue a call with negative sentiment, while measuring sentiment trends over time means you can assess the impact of campaigns, product launches, and agent performance.





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When should I use sentiment analysis? >

Is sentiment analysis hard? >

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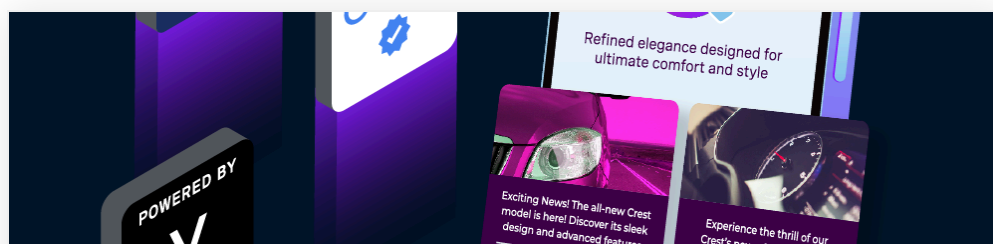
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